cft-anyons Lab Book

From category data to provably rigorous CFTs

cft-anyons contributors

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Contents

1 Frontmatter, North Star, and Claim Status **3**
- 1.1 The Two Success Criteria . . . 3
- 1.2 Claim Status . . . 3
- 1.3 Evidence Chain . . . 3

2 Programme Map **4**
- 2.1 The Pipeline . . . 4
- 2.2 Stage 1 — Category Input . . . 4
- 2.3 Stage 2 — Microscopic Models . . . 4
- 2.4 Stage 3 — Lattice Symmetry Generators . . . 4
- 2.5 Stage 4 — Provable Continuum Limit . . . 4
- 2.6 Stage 5 — Rigorous (C)FT Target . . . 5
- 2.7 First Decisions . . . 5

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar **6**
- 3.1 Sources and Dependencies . . . 6
- 3.2 The Three Primitives . . . 6
- 3.3 Two Monoidal Structures and Their Units . . . 6
- 3.4 The Grammar and its Carrier . . . 7
- 3.5 Particle-Number Grading . . . 7
- 3.6 Lamport Dependency Skeleton . . . 7
- 3.7 Scope and Deferral . . . 8

4 Indefinite Particle Number: the Full Fock Completion **9**
- 4.1 Sources and Dependencies . . . 9
- 4.2 Two Direct Sums — the Load-Bearing Distinction . . . 9
- 4.3 Completed Tensor Powers . . . 9
- 4.4 The Full Fock Space . . . 9
- 4.5 The Grammar Generates the Indefinite-Particle Space . . . 10
- 4.6 Symmetries Come Later (the Deferred Quotient) . . . 10
- 4.7 Lamport Dependency Skeleton . . . 11
- 4.8 Scope and Pitfalls . . . 11

5 A Hilbert-Space Compiler Contract **12**
5.1 Status and Local Anchors . . . 12
5.2 Mini North Star . . . 12
5.3 Input Record . . . 12
5.4 Output Record . . . 12
5.5 Observable-Algebra Policies . . . 13
5.6 Lamport Dependency Skeleton . . . 13
5.7 Boundary . . . 13

6 Symmetry-Decorated Many-Particle Grammar **14**
6.1 Sources and Dependencies . . . 14
6.2 Decorated Atoms . . . 14
6.3 Propagation Through OR and AND . . . 14
6.4 Propagation Through Full Fock . . . 14
6.5 Intertwiners and Symmetry-Preserving Compilation . . . 15
6.6 Lamport Dependency Skeleton . . . 15
6.7 Boundary . . . 15

7 Sectors, Projections, Subspaces, and Quotients **16**
7.1 Sources and Dependencies . . . 16
7.2 The Triangle . . . 16
7.3 Finite-Group Invariants . . . 16
7.4 Coinvariant Quotient . . . 17
7.5 Observable-Algebra Distinctions . . . 17
7.6 Checked Example . . . 17
7.7 Lamport Dependency Skeleton . . . 17
7.8 Boundary . . . 18

8 Exchange Symmetry and Statistics as Sector Selection **19**
8.1 Sources and Dependencies . . . 19
8.2 Two Symmetry Layers . . . 19
8.3 Bose and Fermi Projectors . . . 19
8.4 Quotient Presentations . . . 20
8.5 Para and Anyonic Boundaries . . . 20
8.6 Lamport Dependency Skeleton . . . 20
8.7 Compiler Consequence . . . 20

1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

> Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline

$$\underbrace{\mathcal{C}\ (+\,\text{OPE/conformal data})}_{\text{input}} \longrightarrow \underbrace{\text{microscopic models}}_{\text{lattice Hilbert space + Hamiltonian}} \longrightarrow \underbrace{\text{lattice symmetry generators}}_{\text{e.g. Koo–Saleur } L_n}$$

$$\longrightarrow \underbrace{\text{provable continuum limit}}_{\text{OAR / wavelets}} \longrightarrow \underbrace{\text{rigorous (C)FT}}_{\text{conformal net / VOA / Wightman}}$$

2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see `CONVENTIONS.md`)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo–Saleur lattice Virasoro L_n, $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, a Wightman theory, or an Osterwalder–Schrader theory?
- What is the fidelity statement: that the target's superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite "AND/OR many-particle grammar" developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND $= \otimes$, OR $= \oplus$, $\mathbf{1} = \mathbb{C}$); see that project's shard `AQM-70`. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type's space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum $\oplus$.** "Either a particle of type 1 or of type 2" is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product $\otimes$.** "A particle of type 1 *and* a particle of type 2" is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum = $\mathbb{C}$.** The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §"Symmetries come later"). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\mathrm{Hilb},\ \oplus,\ \{0\}) \qquad \text{and} \qquad (\mathrm{Hilb},\ \otimes,\ \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \ \cong\ A \qquad (\oplus\text{-unit is the } \textit{zero space } \{0\}),$$

$$A \otimes \mathbb{C} \ \cong\ A \qquad (\otimes\text{-unit is the } \textit{vacuum } \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta :\ A \otimes (B \oplus C) \ \xrightarrow{\cong}\ (A \otimes B) \oplus (A \otimes C), \qquad A \otimes \{0\} \ \cong\ \{0\}.$$

Together $(\mathrm{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum $=\mathbb{C}$**, not** $\{0\}$**.** The vacuum is the $\otimes$-unit $\mathbb{C}$ (the empty tensor product, one particle's worth of "nothing"), a *one-dimensional* space carrying the unit vector Ω. It is *not* the $\oplus$-unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x\in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E,F) \mid \mathrm{OR}(E,F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1}=\mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\mathcal{V}(0)=\{0\}, \qquad \mathcal{V}(\mathbf{1})=\mathbb{C}, \qquad \mathcal{V}(K)=K,$$
$$\mathcal{V}(\mathrm{AND}(E,F))=\mathcal{V}(E)\otimes\mathcal{V}(F), \quad \mathcal{V}(\mathrm{OR}(E,F))=\mathcal{V}(E)\oplus\mathcal{V}(F).$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\mathcal{V}_0(\mathbf{1})=\mathbb{C}, \quad \mathcal{V}_n(\mathbf{1})=\{0\}\ (n\neq 0), \qquad \mathcal{V}_1(K)=K, \quad \mathcal{V}_n(K)=\{0\}\ (n\neq 1), \quad \mathcal{V}_n(0)=\{0\},$$
$$\mathcal{V}_n(\mathrm{OR}(E,F))=\mathcal{V}_n(E)\oplus\mathcal{V}_n(F), \qquad \mathcal{V}_n(\mathrm{AND}(E,F))=\bigoplus_{i+j=n}\mathcal{V}_i(E)\otimes\mathcal{V}_j(F).$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E)=\bigoplus_{n\geq 0}\mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle "menu" over a finite type set X is $P=\mathrm{OR}_{x\in X}K_x$, with $\mathcal{V}(P)=\bigoplus_{x\in X}K_x$ (the infinite, ℓ^2-completed menu is CA-03). The expression $\mathbf{1}\,\mathrm{OR}\,K$ has carrier $\mathbb{C}\oplus K$ — a "maybe" particle (either zero or one particle of type K), as in the sibling shard's $\mathbb{C}\oplus\mathbb{C}^d$. Tensoring N "maybe" atoms, $\mathrm{AND}_{j=1}^{N}(\mathbf{1}\,\mathrm{OR}\,K)$, has carrier $(\mathbb{C}\oplus K)^{\otimes N}$, whose grade-$n$ sector (by distributivity) is $\bigoplus_{|S|=n}\bigotimes_{x\in S}K$ over n-subsets of the N slots — the finite "occupation" picture.

3.6 Lamport Dependency Skeleton

$D1$: atoms $\{K_x\}_{x\in X}$ (single-particle Hilbert spaces); $\mathbf{1}=\mathbb{C}$ (vacuum), $0=\{0\}$.
$D2$: $E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E,F) \mid \mathrm{OR}(E,F)$.
$D3$: $\mathcal{V}:\ \mathcal{V}(0)=\{0\},\ \mathcal{V}(\mathbf{1})=\mathbb{C},\ \mathcal{V}(\mathrm{AND})=\otimes,\ \mathcal{V}(\mathrm{OR})=\oplus$.
$D4$: $\mathcal{V}_0(\mathbf{1})=\mathbb{C},\ \mathcal{V}_1(K)=K;\ \mathcal{V}_n(\mathrm{OR})=\oplus,\ \mathcal{V}_n(\mathrm{AND})=\bigoplus_{i+j=n}\mathcal{V}_i\otimes\mathcal{V}_j$.
$D5$: $(\mathrm{Hilb},\oplus,\{0\},\otimes,\mathbb{C})$ a symmetric rig category, with
$\delta : A\otimes(B\oplus C)\xrightarrow{\cong}(A\otimes B)\oplus(A\otimes C)$ unitary and natural.
$T1$: (D5) coherence (Laplaza; Kelly) $\Rightarrow\mathcal{V}$ well defined up to canonical unitary iso.
$T2$: (D4) finite E : $\mathcal{V}(E)=\bigoplus_{n\geq 0}\mathcal{V}_n(E)$ a finite sum (finite particle number).

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$-unit is $\{0\}$; the $\otimes$-unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02's finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project's finite grammar (`AQM-70`) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n\geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\mathrm{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|^2_{W_n} < \infty$, with inner product $\langle (w_n), (v_n) \rangle = \sum_n \langle w_n, v_n \rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\mathrm{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\mathrm{alg}}$. "Indefinite particle number" $\Leftrightarrow$ "not confined to finitely many sectors" $\Rightarrow$ the ℓ^2 completion is *required.* This is exactly what CA-02's finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K, the *completed* (Hilbert) tensor product $H \mathbin{\widehat{\otimes}} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a \otimes b,\, c \otimes d \rangle = \langle a, c \rangle_H \langle b, d \rangle_K$ (well defined and positive definite). For finite-dimensional factors $\widehat{\otimes} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \;\; \text{(the vacuum sector)}, \qquad H^{\widehat{\otimes} n} := \underbrace{H \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} H}_{n},$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H, the *full Fock space* is

$$\boxed{\mathfrak{F}(H) \;:=\; \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}}, \qquad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n\geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum.* This is the *distinguishable* Fock space (the author's term in the local source `1910.10619`); in operator algebra it is also called the *free* or *Boltzmann* Fock space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see "Symmetries come later" below.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\mathrm{alg}} H^{\widehat{\otimes} n}$ (the finite-particle vectors) is *dense*; (ii) distinct-n sectors are *mutually orthogonal*, $\langle\psi,\varphi\rangle = \sum_n \langle\psi_n,\varphi_n\rangle$, so $H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2-direct-sum facts above with $W_n = H^{\widehat{\otimes} n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. $\square$

The $1/N!$ caveat (a convention trap). Schottenloher's inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \mathrm{OR}_{x\in X} K_x$, with carrier $\mathcal{V}(P) = \widehat{\bigoplus}_{x\in X} K_x$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02's δ, extended unitarily to the completions) gives the $(n, \text{type-word})$-graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes} n} \;\cong\; \widehat{\bigoplus_{w\in X^n}} K_w, \quad K_w := K_{x_1} \,\widehat{\otimes} \cdots \widehat{\otimes}\, K_{x_n}, \qquad \Longrightarrow \qquad \mathfrak{F}(\mathcal{V}(P)) \;\cong\; \widehat{\bigoplus_{n\geq 0}}\, \widehat{\bigoplus_{w\in X^n}} K_w.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \widehat{\bigoplus}_{w\in X^n} K_w$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2-completed, all-n analogue of CA-02's finite grade-n sector. The sibling project's finite expressions ($P_X^{\mathrm{AND}\,m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite-M restrictions of this object; that is the exact harmonisation: `AQM-70` is the $\bigoplus^{\mathrm{alg}}$/finite-M truncation of $\mathfrak{F}$.

4.6 Symmetries Come Later (the Deferred Quotient)

The full Fock space is the carrier on which exchange symmetry acts; it is not itself a (anti)symmetric space. For each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes} n}$ by permuting tensor slots. Bosonic and fermionic Fock spaces are then the images of the per-sector orthogonal projectors

$$\Gamma_{\mathrm{s}}(H) = P_{\mathrm{sym}}\,\mathfrak{F}(H) \quad \text{(symmetric / Bose)}, \qquad \Gamma_{\mathrm{a}}(H) = P_{\mathrm{anti}}\,\mathfrak{F}(H) \quad \text{(antisymmetric / Fermi)},$$

with para-statistics the other Young-symmetry isotypic components. This is the "impose an equivalence relation under particle exchange" step. We define no projector and prove nothing here: the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

$D6$: $\bigoplus^{\mathrm{alg}}$(finite support) $\subset$ $\widehat{\bigoplus}(\ell^2)$: $\langle (w_n),(v_n)\rangle = \sum_n \langle w_n, v_n\rangle$.
$D7$: completed tensor power $H^{\widehat{\otimes} n}$, $H^{\widehat{\otimes} 0} = \mathbb{C}$ (vacuum).
$D8$: $\mathfrak{F}(H) := \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}$ (full / distinguishable Fock); $\Omega = (1, 0, \dots)$.
$D9$: $P = \mathrm{OR}_{x\in X} K_x$; $\quad \mathfrak{F}(\mathcal{V}(P)) = \widehat{\bigoplus}_{n\geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}$.
$T3$: (D6,D7) $\mathfrak{F}(H)$ Hilbert; $\bigoplus^{\mathrm{alg}}$ dense; $H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}$ $(m \neq n)$.
$T4$: indefinite particle no. $\Leftrightarrow$ infinite support $\Rightarrow \widehat{\bigoplus}$ (not $\bigoplus^{\mathrm{alg}}$).
$T5$: (D8,δ) $\mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w$ $((n,\text{type-word})$ grading).
$T6$: $\Gamma_{\mathrm{s}} = P_{\mathrm{sym}}\mathfrak{F}$, $\Gamma_{\mathrm{a}} = P_{\mathrm{anti}}\mathfrak{F}$ deferred (statistics later).

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\mathrm{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H, the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2-summable and cross-sector inner products vanish.

5 A Hilbert-Space Compiler Contract

5.1 Status and Local Anchors

Status: Proposal / contract shard. This shard introduces no new mathematical classification theorem. It records a design target for the pre-work sequence around CA-02 and CA-03: a typed grammar of many-particle expressions should be compilable to Hilbert-space data and observable-algebra data with explicit evidence status.

Local anchors are CA-02 for the finite AND/OR/vacuum grammar, CA-03 for the Hilbert direct-sum Fock completion, and `CONVENTIONS.md` entries (f) and (g) for symmetry and sector semantics. The first executable check for this sequence is the finite averaging projector in `test/runtests.jl`, backed by `src/CftAnyons.jl`.

5.2 Mini North Star

Proposal: build a *Hilbert-space compiler*. The user supplies an expression in a typed grammar, together with the atom table and required policies; the compiler returns the Hilbert carrier, its observable algebra policy, its symmetry representation data, and any sector projection or quotient presentation that was requested.

The motivating analogy is "regular expressions for quantum mechanics": a small syntax has compositional semantics. The analogy is only a prompt. Here the syntax is typed and semantics-bearing; every constructor must say what it does to Hilbert carriers, particle gradings, observable algebras, symmetries, and evidence metadata.

5.3 Input Record

A compiled expression is not just a string. It has an input record:

$$\mathsf{Input} = (\mathsf{Atoms},\ E,\ \mathsf{Completion},\ \mathsf{ObsPolicy},\ \mathsf{Symmetry},\ \mathsf{SectorPolicy}).$$

Atoms. A table of atoms. At minimum, each atom names a Hilbert space K_x. Later entries may include a group representation, a particle degree, a source locator, or category data.

E. A grammar expression built from $0, \mathbf{1}, K_x, \mathrm{AND}, \mathrm{OR}$, Fock, and later sector constructors.

$\mathsf{Completion}$. Whether $\oplus, \otimes$ are finite Hilbert operations, Hilbert completions $\widehat{\bigoplus}, \widehat{\otimes}$, or a finite cutoff. CA-03 fixes the first infinite Fock completion used here.

$\mathsf{ObsPolicy}$. The observable-algebra choice. There is no silent default: full $\mathcal{B}(H)$, block-preserving, tensor-local, symmetry-invariant, and compressed sector algebras are different outputs.

$\mathsf{Symmetry}$. Optional unitary or projective-unitary representation data, handled according to CONVENTIONS.md (f).

$\mathsf{SectorPolicy}$. Optional projection, subspace, or quotient data; see CONVENTIONS.md (g).

5.4 Output Record

The intended output is a typed record, not only a Hilbert-space dimension:

$$\mathsf{Compile}(E) = (H_E,\ \{H_{E,n}\}_{n\geq 0},\ \mathcal{A}_E,\ U_E,\ P_E,\ Q_E,\ \mathsf{Evidence}_E).$$

H_E. The Hilbert carrier. CA-02 and CA-03 define the first carrier semantics.

$\{H_{E,n}\}$. The particle-number grading when the atom table declares particle degrees.

$\mathcal{A}_E$. The selected observable algebra. This slot is mandatory once any observable claim is made.

U_E. The propagated symmetry representation, if symmetry data were supplied.

P_E. The selected sector projection, if a sector constructor was used.

Q_E. The quotient presentation, if one was requested and its closed relation subspace was named.

$\mathsf{Evidence}_E$. Source-local, checked, proposal, question, or rejected evidence tags for the pieces above.

5.5 Observable-Algebra Policies

The carrier alone does not determine the observable algebra relevant to a model. For example, $H \oplus K$ supports at least:

$$\mathcal{B}(H \oplus K), \qquad \mathcal{B}(H) \oplus \mathcal{B}(K), \qquad \text{off-diagonal creation/annihilation blocks between } H \text{ and } K.$$

All three are legitimate in different contexts; none is forced by the word "OR" alone. Similarly, $H \otimes K$ supports the full algebra $\mathcal{B}(H \otimes K)$ and the tensor-local subalgebra generated by $\mathcal{B}(H) \otimes I$ and $I \otimes \mathcal{B}(K)$.

Compiler rule: no report statement may infer an observable algebra merely from a carrier constructor. The statement must name the observable policy, or remain a carrier-only statement.

5.6 Lamport Dependency Skeleton

$$\begin{aligned}
D10 &: \mathsf{Input} = (\mathsf{Atoms}, E, \mathsf{Completion}, \mathsf{ObsPolicy}, \mathsf{Symmetry}, \mathsf{SectorPolicy}).\\
D11 &: \mathsf{Compile}(E) = (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Evidence}_E).\\
D12 &: \mathsf{ObsPolicy} \in \{\text{carrier-only}, \mathcal{B}(H), \text{block}, \text{tensor-local}, \text{invariant}, \text{compressed}\}.\\
T7 &: \text{A carrier claim needs } H_E;\ \text{an observable claim also needs } \mathcal{A}_E.\\
T8 &: \text{The compiler analogy is a proposal, not a theorem of canonical quantization.}
\end{aligned}$$

5.7 Boundary

This contract does not yet compile fusion-category labels, anyonic braid statistics, Hamiltonians, continuum limits, or CFT data. It only fixes the pre-work target: future shards should be able to say exactly which input record they compile and which output slots they have actually justified.

6 Symmetry-Decorated Many-Particle Grammar

6.1 Sources and Dependencies

This shard extends CA-02 and CA-03 by decorating atoms with symmetry data. It uses CONVENTIONS.md (f).

6.2 Decorated Atoms

Fix a topological group G. A symmetry-decorated atom is a pair

$$(K_x, U_x), \qquad U_x : G \to U(K_x),$$

where K_x is the atomic Hilbert space and U_x is a unitary representation. If the intended symmetry is projective, the pre-work convention is to choose a central extension

$$1 \to U(1) \to E \to G \to 1$$

and use an honest unitary representation $U_x : E \to U(K_x)$. The expression is then compiled over E, with a note that the induced physical symmetry is projective over G.

The vacuum atom is $(\mathbb{C}, \mathbf{1})$, where $\mathbf{1}(g)z = z$. The zero atom carries the unique representation on $\{0\}$.

6.3 Propagation Through OR and AND

If E, F have compiled carriers and representations (H_E, U_E) and (H_F, U_F) for the same group or chosen central extension, define

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad U_{\mathrm{OR}(E,F)}(g) = U_E(g) \oplus U_F(g),$$

and

$$H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F, \qquad U_{\mathrm{AND}(E,F)}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

For finite-dimensional factors, $\widehat{\otimes} = \otimes$. In infinite-dimensional Hilbert spaces, the completed tensor product is the one already required by CA-03.

These formulas are the Hilbert-space version of the monoidal structure on $\mathrm{Rep}^{\dagger}(G)$: direct sums are block actions and tensor products are diagonal actions. If two atoms use different groups, different central extensions, or incompatible projective multipliers, the compiler must stop and ask for a common symmetry object before forming a single U_E.

6.4 Propagation Through Full Fock

For an ordinary unitary representation $U : G \to U(H)$, define the full-Fock action

$$\mathfrak{F}(U)(g) = \bigoplus_{n \geq 0} U(g)^{\widehat{\otimes} n} \quad \text{on} \quad \mathfrak{F}(H) = \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n},$$

with $U(g)^{\widehat{\otimes} 0} = \mathrm{id}_{\mathbb{C}}$. This preserves the particle-number grading. In this pre-work shard, no Shale–Stinespring-type implementability theorem is claimed for general canonical field algebras; the formula is only the sectorwise tensor-power representation on the full Fock carrier. Implementing a one-particle unitary on a chosen CAR/CCR observable algebra is a separate field-algebra question.

6.5 Intertwiners and Symmetry-Preserving Compilation

A morphism between decorated atoms or compiled expressions is symmetry-preserving only when it is an intertwiner:

$$TU_E(g) = U_F(g)T \quad \text{for all } g.$$

This matters for compiler optimisations. Rebracketing and distributivity isomorphisms used by CA-02 are acceptable in the decorated grammar only when they also intertwine the propagated actions. The standard Hilbert-space associators, unitors, symmetries, and distributivity maps do so for diagonal direct-sum and tensor-product actions.

6.6 Lamport Dependency Skeleton

$$\begin{aligned}
D13 &: \text{decorated atom } (K_x, U_x),\ U_x : G \to U(K_x).\\
D14 &: \text{projective symmetry over } G \text{ is compiled via } U_x : E \to U(K_x) \text{ for a named central extension } E.\\
D15 &: U_{E\,\mathrm{OR}\,F} = U_E \oplus U_F, \quad U_{E\,\mathrm{AND}\,F} = U_E \mathbin{\widehat{\otimes}} U_F.\\
D16 &: \mathfrak{F}(U) = \widehat{\bigoplus}_{n\geq 0} U^{\widehat{\otimes} n}, \quad U^{\widehat{\otimes} 0} = 1_{\mathbb{C}}.\\
T9 &: \text{The propagated symmetry preserves the CA-02/CA-03 particle grading.}\\
T10 &: \text{A single compiled } U_E \text{ requires a common group or central extension.}
\end{aligned}$$

6.7 Boundary

This shard does not yet impose invariance, select charge sectors, form quotient spaces, or define symmetry-invariant observable algebras. Those are sector operations, not mere symmetry propagation, and are treated in CA-06.

7 Sectors, Projections, Subspaces, and Quotients

7.1 Sources and Dependencies

This shard depends on CA-04 and CA-05 and fixes the first sector semantics for the Hilbert-space compiler. It uses CONVENTIONS.md (g).

7.2 The Triangle

Let H be a Hilbert space. The compiler treats a sector first as an orthogonal projection

$$P = P^* = P^2 \in \mathcal{B}(H).$$

It may then expose either of the equivalent Hilbert-space presentations

$$H_P := \operatorname{Ran}(P) \subset H, \qquad H/\ker(P) = H/\operatorname{Ran}(1-P).$$

The map

$$[v] \in H/\ker(P) \longmapsto Pv \in \operatorname{Ran}(P)$$

is a well-defined unitary isomorphism when the quotient has the Hilbert quotient norm. Thus the safe slogan is not "symmetry equals quotient" but:

$$\text{projection} \quad \Longleftrightarrow \quad \text{closed subspace} \quad \Longleftrightarrow \quad \text{quotient by the named orthogonal complement.}$$

7.3 Finite-Group Invariants

Let G be finite and $U : G \to U(H)$ a unitary representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} U(g).$$

Then P_G is the orthogonal projection onto the invariant subspace

$$H^G = \{v \in H : U(g)v = v \ \forall g \in G\}.$$

Proof. Idempotence follows by reindexing:

$$P_G^2 = \frac{1}{|G|^2} \sum_{g,h} U(gh) = \frac{1}{|G|} \sum_{k} U(k) = P_G.$$

Self-adjointness follows from unitarity and $g \mapsto g^{-1}$:

$$P_G^* = \frac{1}{|G|} \sum_{g} U(g)^* = \frac{1}{|G|} \sum_{g} U(g^{-1}) = P_G.$$

For every k, $U(k)P_G = P_G$, so $\operatorname{Ran}(P_G) \subset H^G$. If $v \in H^G$, then $P_G v = v$, so $H^G \subset \operatorname{Ran}(P_G)$. $\square$

7.4 Coinvariant Quotient

For the same finite action, let

$$R_G = \operatorname{span}\{U(g)v - v : g \in G,\ v \in H\}.$$

Then $R_G \subseteq \ker(P_G)$ because $P_G U(g) = P_G$. Conversely, if $w \in \ker(P_G)$, then

$$w = \frac{1}{|G|} \sum_{g \in G} \bigl(w - U(g)w\bigr) \in R_G.$$

Therefore $R_G = \ker(P_G) = (H^G)^\perp$, and the coinvariant quotient

$$H_G := H/R_G$$

is canonically unitarily isomorphic to the invariant subspace H^G. For compact non-finite groups the analogous statement requires a Haar-integral and analytic hypotheses; this shard records only the finite checked case.

7.5 Observable-Algebra Distinctions

There are three different symmetry-related algebras that the compiler must not identify silently:

$$\mathcal{B}(H), \qquad \mathcal{B}(H)^G := \{A : U(g)AU(g)^* = A\ \forall g\}, \qquad P_G \mathcal{B}(H) P_G \cong \mathcal{B}(H^G).$$

The first is the full algebra before imposing symmetry. The second is the algebra of invariant observables on the original carrier. The third is the compressed algebra acting on the invariant sector. A model may choose one of these, but the choice is an observable policy in the sense of CA-04, not a consequence of the word "symmetry" alone.

7.6 Checked Example

The current Julia check uses $G = C_2$ acting on $\mathbb{C}^2$ by the swap matrix S. The averaging projection is

$$P = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

The test checks $P^* = P = P^2$, checks $I - P$ is the complementary orthogonal projection, verifies $P(I - P) = 0$, and verifies that $(1,1)$ is fixed while $(1,-1)$ is killed. This is the first executable invariant for the sector compiler.

7.7 Lamport Dependency Skeleton

$D17$: sector projection $P = P^* = P^2$; $\quad H_P = \operatorname{Ran}(P)$.
$D18$: $H/\ker(P) \xrightarrow{[v] \mapsto Pv} \operatorname{Ran}(P)$.
$D19$: $P_G = |G|^{-1} \sum_g U(g) \quad$ (G finite).
$D20$: $R_G = \operatorname{span}\{U(g)v - v\}$.
$L1$: $P_G = P_G^* = P_G^2, \quad \operatorname{Ran}(P_G) = H^G$.
$L2$: $R_G = \ker(P_G) = (H^G)^\perp, \quad H/R_G \cong H^G$.
$T11$: Projection, subspace, and quotient are equivalent only after the relation subspace is named.
$T12$: $\mathcal{B}(H)^G$ and $P_G \mathcal{B}(H) P_G$ are different observable policies.

7.8 Boundary

The finite averaging theorem is not a statement about spontaneous symmetry breaking, gauge constraints, BRST cohomology, braid statistics, or superselection sectors in a continuum CFT. It is the first Hilbert-space sector rule needed by the compiler.

8 Exchange Symmetry and Statistics as Sector Selection

8.1 Sources and Dependencies

This shard discharges part of the deferral in CA-03: ordinary Bose/Fermi statistics are treated as sector selections on the full distinguishable Fock carrier. It depends on CA-05 for internal symmetry propagation and CA-06 for projection/subspace/quotient semantics.

8.2 Two Symmetry Layers

There are two distinct actions on many-particle carriers:

$$\text{internal symmetry } G, \qquad \text{exchange symmetry } S_n.$$

If $U : G \to U(H)$ is an internal one-particle representation, then on the n-particle distinguishable sector $H^{\widehat{\otimes} n}$ it acts diagonally:

$$U_n(g) = U(g)^{\widehat{\otimes} n}.$$

The permutation group S_n acts by permuting tensor slots:

$$\Pi_n(\sigma)(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The two actions commute:

$$\Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).$$

Thus internal charge sectors and exchange-statistics sectors can be imposed in either order in the ordinary tensor-slot setting.

8.3 Bose and Fermi Projectors

For finite S_n, CA-06 gives averaging projectors. The symmetric projector is

$$P_{\mathrm{s},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_n(\sigma),$$

and the antisymmetric projector is

$$P_{\mathrm{a},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\Pi_n(\sigma).$$

Their ranges are the symmetric and antisymmetric tensor powers:

$$\operatorname{Ran}(P_{\mathrm{s},n}) = H^{\widehat{\otimes}_s n}, \qquad \operatorname{Ran}(P_{\mathrm{a},n}) = \bigwedge^n H.$$

The corresponding Fock sectors are

$$\Gamma_{\mathrm{s}}(H) = \widehat{\bigoplus}_{n\geq 0} \operatorname{Ran}(P_{\mathrm{s},n}), \qquad \Gamma_{\mathrm{a}}(H) = \widehat{\bigoplus}_{n\geq 0} \operatorname{Ran}(P_{\mathrm{a},n}).$$

This makes precise the CA-03 statement that full/distinguishable Fock is the carrier and Bose/Fermi spaces are later sector choices.

8.4 Quotient Presentations

The symmetric sector can also be presented as a quotient of the full tensor algebra by the closed linear span of relations

$$v_1 \otimes \cdots \otimes v_n - v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The antisymmetric sector uses the signed relations

$$v_1 \otimes \cdots \otimes v_n - \mathrm{sgn}(\sigma) v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

In the compiler, these quotient presentations are secondary: the primary sector object is the orthogonal projector, because it fixes the inherited Hilbert structure and the compressed observable algebra.

8.5 Para and Anyonic Boundaries

Para-statistics would require selecting other irreducible S_n-isotypic components, hence central idempotents or matrix-block projectors for the group algebra of S_n. This shard records that as a proposal, not as a completed construction; a later shard must source the representation convention and the projector normalization before using it.

Genuinely anyonic exchange is not an S_n-projection problem on ordinary tensor slots. It involves braid-group or categorical data and, in this project, must be handled together with the fusion-category conventions for associators, braiding, and F/R-symbols. No braid-statistics projector is defined here.

8.6 Lamport Dependency Skeleton

$$\begin{aligned}
D21 &: U_n(g) = U(g)^{\widehat{\otimes} n} \quad \text{(internal symmetry on } H^{\widehat{\otimes} n}).\\
D22 &: \Pi_n : S_n \to U(H^{\widehat{\otimes} n}) \quad \text{permutes tensor slots.}\\
D23 &: P_{\mathrm{s},n} = n!^{-1} \textstyle\sum_\sigma \Pi_n(\sigma), \quad P_{\mathrm{a},n} = n!^{-1} \textstyle\sum_\sigma \mathrm{sgn}(\sigma)\Pi_n(\sigma).\\
L3 &: \Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).\\
L4 &: P_{\mathrm{s},n}, P_{\mathrm{a},n} \text{ are orthogonal projections by CA-06 finite averaging.}\\
T13 &: \Gamma_\mathrm{s}, \Gamma_\mathrm{a} \text{ are sector selections inside full Fock.}\\
T14 &: \text{Anyonic exchange remains deferred to categorical braid data.}
\end{aligned}$$

8.7 Compiler Consequence

The compiler now has its first nontrivial sector family:

$$\mathsf{Compile}(\mathrm{Bose}(\mathfrak{F}(H))) = (\Gamma_\mathrm{s}(H),\ P_\mathrm{s},\ \text{chosen compressed algebra}, \ldots)$$

and similarly for Fermi. The sector constructor is separate from Fock itself, preserving the separation of indefinite particle number from exchange symmetry.